Reliable Self-assembly by Self-triggered Activation of Enveloped DNA Tiles

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Abstract. Although the design of DNA tiles has been optimised for efficient and specific self-assembly, assembly errors occur so often that applications for molecular computation remain limited. We propose the use of an enveloped tile consisting of a DX- base tile that carries a protector tile to suppress erroneous tile assembly. The design of the enveloped tile promotes the dissociation of the protector tile from the base tile through a self-triggered activation process, which keeps the outputs of the base tile stay protected until both base tile inputs have bonded correctly to the assembly. The enveloped tile design, the self-triggered activation that removes the protector tile and preliminary modelling results are presented.

Keywords: Enveloped Tile, Self-assembly, DNA tile, Molecular computation, Protector Tile.

1 Introduction and Motivation

DNA tile self-assembly [9] — the emergence of assemblies through physicochemical interactions between basic building blocks termed *DNA tiles*, provides a potential technology for programmable computation at the molecular scale. DNA tiles are abstract representations of DNA molecules, e.g. DNA double crossover (DX) molecule [11], with certain number of protruding sticky ends of single stranded DNAs (ssDNAs). DNA tiles may be designed with combinatorial DNA sticky ends that can bind together by canonical base pairing (A-T and G-C binding), and thereby tiles may self-assemble. However, the DNA tile assembly process is challenged by a high level of assembly errors.

There are three basic types of assembly errors: growth errors [10], facet nucleation errors [3], and spontaneous nucleation errors [8]. The first two types of errors both result from erroneous tile trapping. Growth errors occur when a tile gets trapped at a vacant site of a growing assembly despite one of its sticky

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ends failing to match correctly. Facet nucleation errors appear when an incorrectly attached tile at a growing surface fails to detach quickly and eventually gets fixed. Spontaneous nucleation errors are attributed to spurious assemblies growing without a seed.

Tile designs for error prevention and correction include: redundant tile sets [5,10], protected tiles [4,6] and morphological encoding [2,12]. In redundant tile sets, each tile is replaced by a kxk tile set e.g. proofreading tiles [10] and snake proofreading tiles [5]. However, the improved error resilience requires k^2 times more tiles. Such a solution is not only disadvantaged by the increase in the necessary resource but further designing larger tile sets with orthogonal sticky ends imposes practical challenges.

The protected tile approach avoids such resource increases e.g. Protected and Layered Tiles [4] and "Activatable" Tiles [6]. However, the Protected and Layered tiles are not fully protected on their outputs — only three bases in the best case and leaving 11 bases vulnerable for spurious binding. On the other hand, the inputs and outputs of the Activatable tiles are fully protected. However, the approach requires the support of biological enzymes, which poses practical limitation.

The third approach uses morphological encoding to enhance specificity of binding i.e. correct binding between two tiles requires a match of both the binding patterns and the geometry of the involved inputs and outputs [2,12]. However, such morphological encoding poses a further design challenge due to the need for more than one sticky end at each of the inputs and outputs.

In this work, the Base Tile design remains similar to that of the well established DX-tile [11]. The assembly error resilience is thus achieved through the new Protector Tile in combination with the Base Tile termed Enveloped Tile. In particular, the design of the Protector Tile controls the self-triggered activation process resulting in either the correct attachment of the Base Tile to the assembly or release of it (if erroneous) and thus ensures directional growth of the assembly.

The remainder of the article is structured as follows: background of DNA tile self-assembly and its modelling are described in section 2. Section 3 presents proposed Enveloped Tile, and further describes design of Base Tiles and Protector Tiles for Sierpinski Triangle assembly [10]. In Section 4, self-assembly steps involving association of Enveloped Tile and self-triggered dissociation of Protector Tile has been discussed. Further, in section 5, error prevention performance of Enveloped Tile assembly has been analysed by a physically realistic kinetic model and preliminary results are documented. Section 6, concludes the article.

2 Background

In order to assemble uniquely defined structures or to perform a computation step using DNA tile self-assembly, a certain tile set with properly designed sticky ends is required. The tile set often consists of a seed tile, boundary tiles and rule tiles [10]. Binding strengths of sticky ends of these tiles depend on their respective ssDNAs. Two such tiles may join together if their sticky ends are complementary and of the same length. However, such binding is a reversible chemical process, and an assembled tile is stable only if its associated total binding strength is sufficient to prevent its dissociation under local physical conditions.

The kinetic Tile Assembly Model (kTAM) [10] provides a way for mathematical modelling of the proposed tile solution. The kTAM considers each tile self-assembly step as a reversible process governed by the tile concentration, local reaction temperature and the length of the tile's sticky ends. Furthermore, it assumes that a) the tile concentration is constant for each tile type and b) only one tile can attach/detach from the growing aggregate at a time. The model enables analysis of the assembly errors and growth rate for a given tile set. Further, the tile design itself may be optimised through parameter adjustment.

Reversible kinetics of tile assembly leading to association or dissociation from a vacant site of the growing aggregate can be represented by equivalent forward and reverse reaction rates, respectively. The rate of tile attachment at a binding site of an aggregate is directly proportional to the tile concentration. The concentration of each type of tile (except the seed tile) can be given by $e^{-G_{mc}}$, where G_{mc} is the decrease in entropy when a tile binds at a vacant site. Therefore, the forward reaction rate (r_f) can be given by $r_f = k_f e^{-G_{mc}}$ where k_f is the reaction rate constant. Similarly, the tile detachment process is controlled by the energy required to break any single tile-aggregate bond and denoted by G_{se}. The value of G_{se} depends on the sticky end length (s) and the temperature (T), where $G_{se} \approx (4000/T - 11)s$. The tile reverse reaction rate involving b tile bonds is given by $r_{r,b} = k_f e^{-bG_{se}}$.

A larger value of G_{mc} thus implies a lower tile concentration and consequently a slower forward reaction rate (or vice versa). Similarly, a larger value of G_{se} results in a slower detachment rate. The optimum growth rate with low error rates happens near thermodynamic equilibrium ($G_{mc} \approx 2G_{se}$) [10], and may be given by $r_g \approx \frac{r_f - r_{r,2}}{2}$ and $\varepsilon \approx e^{-G_{se}}$, respectively. Therefore, a relation between optimum growth rate and minimum error rate may be given by $r_g \approx \beta\varepsilon^2$ where $\beta = 10^5$ /M/sec. Thus, any effort to reduce the error rate (ε) by tuning physical parameters (G_{mc} and G_{se}) would result in a quadratic reduction of the growth rate. However, error reduction without significant fall off in assembly growth rate has been achieved by adding redundant tiles [10,5] and by protecting tile's inputs and outputs [4,6].

3 Proposed *Enveloped Tile* Design

The Base Tile (*BT*) of the Enveloped Tile (*ET*) is designed based on the DNA double crossover Anti-parallel molecule with Odd spacing (DAO molecule) [11] — an accepted building block for DNA tile self-assembly. A Protector Tile (*PT*) provides the unique tile design, assembling with the *BT* and thus protecting the *BT's* inputs and outputs whilst leaving protruding DNA toeholds for self-triggered activation of *BT*.

3.1 Enveloped Tile

The centerpiece of the proposed design is the *Protector Tile* — illustrated in Figure 1(a), involving four ssDNAs (DS-1, DS-2, DS-3 and DS-4). DS-1 and DS-2 serve as protection for the *BT's* input and output sticky ends, respectively, and are clamped together by DS-3 and DS-4. The double stranded DNA (dsDNA) sections on both sides of the central clamp act as distal clamps, linking the proofreading events at the input side to the strand displacement events at the output side.

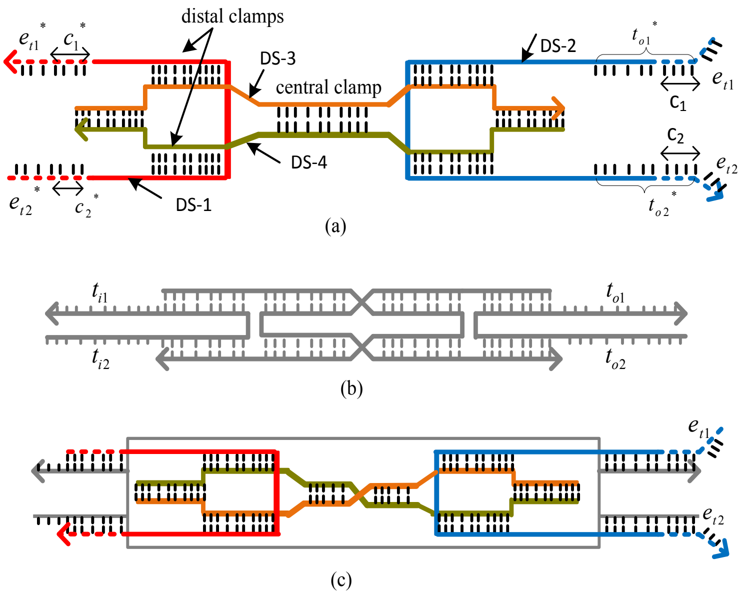

Fig. 1. (a) Protector Tile (*PT*) (b) Base Tile (*BT*) and (c) Enveloped Tile(*ET*)

Each dsDNA section (depicted by vertical dashed lines) is 10 nucleotides (*nt*) long and formed by two anti-parallel ssDNAs. One side of each ssDNA is marked by an arrowhead representing its 3' end, the other side of it is the 5' end. The 7 *nt* sticky ends of DS-1 (e^*_{t1}, c^*_1 and e^*_{t2}, c^*_2) are complementary to the 7 *nt* sections (e_{t1}, c_1 and e_{t2}, c_2), respectively. The input sticky ends of the *PT* ((e^*_{t1}, c^*_1 and e^*_{t2}, c^*_2) of DS-1) are also complementary to the 7 nt of the input sticky ends of the *BT* — see Figure 1(c). Similarly, 7nt of the output sticky ends of the *PT* (t^*_{o1} and t^*_2 of DS-2) are complementary to the output sticky ends of the *BT*.

The central clamp in the *PT* keeps its complementary input and output free ends apart. Based on the physics of stiffness and coiling of dsDNA [1], it is assumed that the stiffness of a 10 *nt* dsDNA clamp will favour *PT* binding with the *BT* when the *PT* and the *BT* are allowed to bond. Such bonding occurs prior to the *ET* being made available for the assembly process. Another important aspect of this clamp is that it will be twisted (coiled) by one turn when the *PT* is bound with the *BT* due to the 3' and 5' orientation of its input sticky ends, shown in Figure 1(c).

The two distal clamps play an important role in the case where only one side of the *PT* gets dissociated i.e. an erroneous tile, and by using this process no erroneous tile Assembly happens. The clamps hold the dissociated input ends close to their binding sites (emulating a locally elevated concentration of these binding sites). This in turn prevents the inputs of the *PT* from engaging with its outputs. The clamps, therefore, reinforce the reverse branch migration so that a partially displaced *PT* can return to its original structure when released from the assembly.

The Base Tile (DAO molecule) — see Figure 1(b), has 10 *nt* sticky ends (t_{i1}, t_{i1}, t_{o1} and t_{o2}). Ten *nt* was selected so as to provide sufficient strength for *PT* binding whilst leaving 3 *nt* toeholds on the input side. The crossover points within the *BT* are separated by 16 *nt* (3 half-turns).

3.2 Enveloped Tile Set for the Sierpinski Triangle

Figure 2 illustrates a Sierpinski tile set design using Enveloped Tiles. As shown in (a), each Sierpinski Rule Tile (SRT-00, SRT-11, SRT-01 and SRT-10) has a different combination of input and output sticky ends (complementary sticky ends are marked by *). The *BTs* are shown in (b) and the *PTs* are shown in (c). The Input and output sticky ends of the BTs and TEs are represented by (S^{jk}_{bi1}, S^{jk}_{bi2}),(S^{jk}_{bo1}, S^{jk}_{bo2}), and (S^{jk}_{ei1}, S^{jk}_{ei2}), (S^{jk}_{eo1}, S^{jk}_{eo2}) respectively, where $jk = 00, 11, 01, 10$ represents the type of Rule tile.

There are three requirements for complementary DNA sequences among these sticky end sequences — between the *BT's* inputs and outputs; between the *BTs'* and *PTs'* inputs and outputs and between the PTs' inputs and outputs. Thus

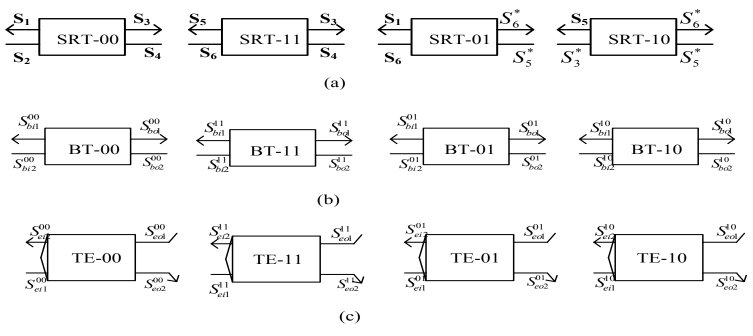

Fig. 2. Enveloped Tile Set for Sierpinski Triangle: (a) Rule Tiles, (b)Base Tiles used for Sierpinski Enveloped Tile set (c) Corresponding Protector Tiles

Table 1. BTs Sticky Ends

Name	DNA sequence
S_{bi1}^{00}	**5'-TCCTAGGACT-3'**
S_{bi2}^{00}	**3'-CTGTGTCAGG-5'**
S_{bo1}^{00}	**3'-AGGTACCTGA-5'**
S_{bo2}^{00}	**5'-GACCAAGTCC-3'**
S_{bi1}^{11}	**5'-CGTTAGGACT-3'**
S_{bi2}^{11}	**3'-GTTTGTCAGG-5'**
S_{bo1}^{11}	**3'-AGGTACCTGA-5'**
S_{bo2}^{11}	**5'-GACCAAGTCC-3'**
S_{bi1}^{01}	**5'-CGTTAGGACT-3'**
S_{bi2}^{01}	**3'-CTGGTTCAGG-5'**
S_{bo1}^{01}	**3'-GCAATCCTGA-5'**
S_{bo2}^{01}	**5'-CAGACAGTCC-3'**
S_{bi1}^{10}	**5'-TCCTAGGACT-3'**
S_{bi2}^{10}	**3'-GTCTGTCAGG-5'**
S_{bo1}^{10}	**3'-GCAATCCTGA-5'**
S_{bo2}^{10}	**5'-CAGACAGTCC-3'**

Table 2. PTs Sticky Ends

Name	DNA sequence
S_{ei1}^{00}	**3'-ACAGTCC-5'**
S_{ei2}^{00}	**5'-ATCCTGA-3'**
S_{eo1}^{00}	**5'-TCCATGGACTGT-3'**
S_{eo2}^{00}	**3'-CTGGTTCAGGAT-5'**
S_{ei1}^{11}	**3'-ATCCTGA-5'**
S_{ei2}^{11}	**5'-ACAGTCC-3'**
S_{eo1}^{11}	**5'-TCCATGGACTGT-3'**
S_{eo2}^{11}	**3'-CTGGTTCAGGAT-5'**
S_{ei1}^{01}	**3'-ATCCTGA-5'**
S_{ei2}^{01}	**5'-CAAGTCC-3'**
S_{eo1}^{01}	**5'-CGTTAGGACTTG-3'**
S_{eo2}^{01}	**3'-GTCTGTCAGGAT-5'**
S_{ei1}^{10}	**3'-ATCCTGA-5'**
S_{ei2}^{10}	**5'-CAAGTCC-3'**
S_{eo1}^{10}	**5'-CGTTAGGACTTG-3'**
S_{eo2}^{10}	**3'-GTCTGTCAGGAT-5'**

the input and output sequences of the PTs are constrained to have a certain sequence of common bases c_1, c_2, shown in Figure 1(a).

The design of the length of the sticky ends of the PT has to take into account a number of factors. These ends must both be able to protect the rule tile's outputs and upon disengaging from the input sides to bond with their input protector counterparts. Therefore, the overhangs have been designed such that the protective helix on the rule tile outputs measures 9 base pairs. Together with the toeholds, the same strands are able to form 7 *nt* bonds internally so as to let the PT dissociate from BT. The DNA sequences c_1 and c_2 are common for all the PTs needed for a Sierpinski tile set. Furthermore, the DNA sequences $(c_1 + e_{t1}, c_2 + e_{t2})$ are 7 *nt* long, therefore adding 3 bases for protruding toeholds on the output side of ETs.

The DNA Design MATLAB scripts [11], were applied to optimise the DNA sequences for BTs and corresponding PTs for the Sierpinski tile set shown in tables 1 and 2. Due to space constraints, only the sticky end sequences of the BTs and PTs are given.

4 Self-assembly Guided by Enveloped Tiles

As illustrated in Figure 3, when a tile approaches a vacant site (I) on the tile assembly, proofreading of the inputs of its BT occurs. Such proofreading results in either a match or no match, between the incoming tile and the vacant site. In case of a match, there are 2 potential cases: case 1 — both inputs match correctly (left half of Figure 3) or case 2 — only one input matches correctly (right half of Figure 3).

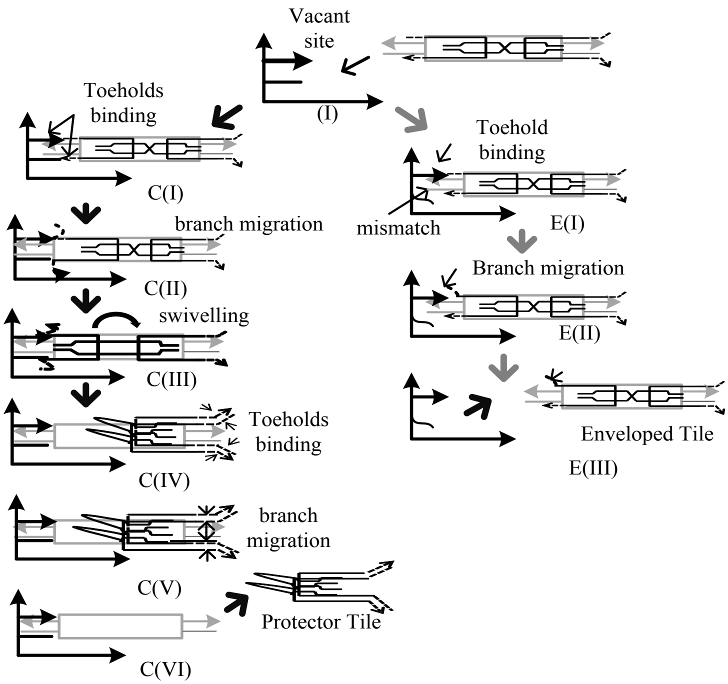

Fig. 3. self-assembly step states: (left) both inputs match, (right) only one input matches

In case 1, the first state – proofreading of the inputs, *C(I)* consists of toehold binding followed by branch migration of 7 *nt* of the inputs of the *PT*, *C(II)*, that were bound with a Rule Tile's sticky ends. A stable binding of the Rule Tile to the assembly has thus occurred. At this stage the Rule Tile no longer needs to be protected by the *PT*. As the *PT* is released from its input side it will result in uncoiling of the clamp, *C(III)*, which primes a configuration in which the left part of the *PT* is free to swivel back towards its outputs where it is still protecting the output sticky ends of the *BT*. The released inputs start 'sniffing' the protruding toeholds at the outputs of the *PT*, *C(IV)*. The resulting toehold binding, followed by a further branch migration, *C(V)*, of the respective 4 *nt* will create a single thermodynamic component to disengage the *PT* and expose the *BT's* output sticky ends, *C(VI)*. The next incoming, matching Rule Tile, provides additional thermodynamic impetus to further support migration of the *PT* away from the growing assembly.

The first stage of Case 2 is as in Case 1, except that only one of the toeholds binds, *E(I)*, and initiates branch migration at that input. All other sticky ends of the *BT* will remain protected, *E(II)*. As this will only occur on one of the input sides all other sticky ends of the non-matching tile will remain protected by the

PT and thus remain unbound to the assembly. Although the branch migration of *E(II)*, may or may not proceed to completion, the otherwise fully protected tile will eventually disengage from the lattice due to the inherently unstable nature of binding through a single input, *E(III)*. The uniqueness of this approach lies in the strategy applied to protect BT's output. The protection comes not just in the form of the binding the outputs but further the distal clamps design protects the outputs from unbinding unless the tile is a correct tile.

5 Performance Analysis

Figure 4 illustrates a continuous time Markov state model of the enveloped tile assembly process. The sequences of states $(MS_{c0}, MS_{c1}, MS_{c2}, MS_{c3}, MS_{c4})$ and $(MS_{e0}, MS_{e1}, MS_{e2})$ represent the assembly process of a correct tile and erroneous tile, respectively. These states are equivalent to the assembly steps ($C(I)$, $C(II)$, $C(IV)$, $C(V)$, $C(VI)$) and ($E(I), E(II), E(III)$), respectively, as shown in Figure 3. The states TC and TE represent the final states of the corresponding Markov chains where either a correct or an erroneous tile gets trapped.

The state transition ($VS \rightarrow MS_{c0}$) represents binding between two 3 *nt* long DNA toeholds of a correctly matching ET with a vacant site. This has been modelled as a reversible process having forward kinetic rate ($r_{fi} = k_{fh}\, e^{-Gmc}$) and reverse kinetic rate ($r_{r,t_{ti2}} = k_{fh}\, e^{-\frac{2\times3}{10}Gse}$) where $k_{fh} = 4 \times 10^6/M/s^2$ [9] is a kinetic rate constant for DNA hybridisation. In the case of ET having single toehold matching with a vacant site, the state transition ($VS \rightarrow MS_{e0}$), two such tiles can engage with the vacant site simultaneously. Thus giving an equivalent forward and reverse rate of $2r_{fi} = 2k_{fh}\, e^{-Gmc}$ and $r_{r,t_{ti1}} = k_{fh}\, e^{-\frac{3}{10}Gse}$, respectively.

The next step involves branch migration of the *PT's* inputs, where the *PT* disengages from the *BT's* input side ($MS_{c0} \rightarrow MS_{c1}$). In the case of single toehold matching, branch migration may still start at this input of *PT* but it is not thermodynamically favourable due to the stiffness of the distal clamps reinforcing the *PT's* return to its bind state ($MS_{e0} \rightarrow MS_{e1}$). The aforementioned branch migration step has been modelled as a reversible process [7] where a 'breaking-forming' base pairing process ideally has no preferred direction. The time required is $\approx 10\,\mu S/$ base-pair, which gives an equivalent kinetic rate $k_{bmi} \approx 10^3$/s in both directions. However, to include a bias in the branch migration processes involving correct and erroneous tiles, two additional dimensionless parameters E_{dc} and E_{de} ($E_{dc} = \frac{E_{de}}{10}$) have been introduced. The new reverse kinetic rates (not shown in Figure 4) are $k_{bmi}e^{-E_{dc}}$ and $k_{bmi}e^{E_{de}}$, reflecting a correct tile and erroneous tile, respectively. Therefore, a larger value of E_{dc} would facilitate release of PT from its input side when a correct ET binds however in the case of erroneous ET binding it would resist the release of PT resulting in better error prevention. For simulation purposes, $E_{dc} = 10$ is chosen.

The next step in the addition of a correct tile ($MS_{c1} \rightarrow MS_{c2}$) represents the binding between the toeholds on the output sides of a *PT* and its own input sticky ends that have just been released. As described above, following the release of

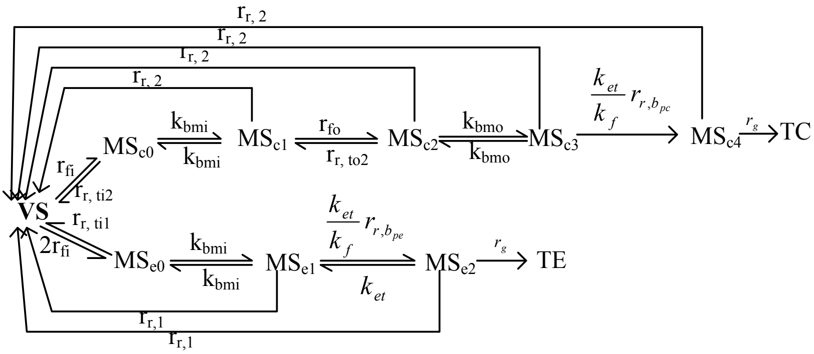

Fig. 4. Kinetic flow graph of Enveloped Tile assembly

the inputs and prior to engaging with its own output strands, the PT undergoes a conformational change by swivelling around its central clamp. It seems likely to assume that this happens instantaneously because the dimensions of a PT are in the range of 10 nm. Therefore, this step has been modelled as a reversible process of DNA hybridisation having a forward rate ($r_{fo} = k_{fh}\, e^{-Gmc}$) and a reverse rate ($r_{r,t_{to2}} = k_{fh}\, e^{-\frac{2\times 3}{10}Gse}$).

The following step ($MS_{c2} \rightarrow MS_{c3}$) represents the branch migration of complementary sections following toeholds on the output ends of the PT. This has been modelled as an unbiased reversible process with kinetic rates of $K_{bmo} = 10^3$/s. Finally, the PT dissociates from its BT ($MS_{c3} \rightarrow MS_{c4}$), modelled as having a unidirectional kinetic rate of dissociation $\frac{k_e t}{k_f h} r_{r,b_{pc}}$ where $r_{r,b_{pc}} = k_{fh} e^{-b_{pc}G_{se}}$ and $b_{pc} = 1$. The b_{pc} represents the total strength with which a PT remains still attached with the BT after its input ends and output ends bind together. The k_{et} represents the effective kinetic rate of localisation of the PT within a BT, which together occupy a volume of $10^3\ nm^3$ and therefore, gives an 'effective concentration' $C_t = \frac{1}{10^3\ nm^3}$ M. Therefore $k_{et} = k_{fh}C_t \approx 6 \times 10^4$. The PT dissociation step in the case of an erroneous tile ($MS_{e1} \rightarrow MS_{e2}$) has parameters similar to a correct tile except that $r_{r,b_{pe}} = k_{fh}e^{-b_{pe}G_{se}}$ and $b_{pe} = 27/10$, which would lead to very slow rates of dissociation under given experimental conditions. The kinetic rate r_g of the final steps ($MS_{c4} \rightarrow TC$ and $MS_{e2} \rightarrow TE$) represents equivalent kinetic rate of tile trapping.

If $p_i(t)$ denotes the probability of the aggregate-tile complex being in state*(i)* at a time t then the self-assembly kinetic rate flow graph of Figure 4 can be written in the form of a Matrix differential equation $\dot{p}(t) = Mp(t)$ where *p(t)* is a 11×1 matrix having $p_i(t)$ (i represents corresponding Markov state) as its elements and

$$M=\begin{bmatrix}
-3r_{fi} & r_{r,ti2} & r_{r,2} & r_{r,2} & r_{r,2} & r_{r,2} & r_{r,1} & r_{r,1} & 0 & 0 & 0 \\
r_{fi} & -k_{bmi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & k_{bmi} & -(r_{r,2}+k_{bmi}+r_{fo}) & r_{r,to2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & r_{fo} & -(r_{r,2}+k_{bmo}+r_{r,to2}) & -(r_{r,2}+r_t) & k_{bmo} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & k_{bmo} & -(r_{r,2}+k_{bmo}+\frac{k_{et}}{k_f}r_{r,b_{pc}}) & k_{et} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{k_{et}}{k_f}r_{r,b_{pc}} & -(r_{r,2}+k_{et}+r_g) & 0 & 0 & 0 & 0 & 0 \\
2r_{fi} & 0 & 0 & 0 & 0 & 0 & -(r_{r,ti1}+k_{bmi}) & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & k_{bmi} & -(r_{r,1}+k_{bmi}+\frac{k_{et}}{k_f}r_{r,b_{pe}}) & r_{et} & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{k_{et}}{k_f}r_{r,b_{pe}} & -(r_{r,1}+k_{et}+r_g) & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & r_g & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & r_g & 0 & 0
\end{bmatrix}$$

The above matrix differential equation has standard solution as $p(t) = e^{Mt}p(0)$ where $p(0)=\left[1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\right]^T$, and another observation that every tile would get fixed either correctly or erroneously if self-assembly is allowed to continue for longer time, gives $\dot{p}(\infty) = \left[1\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ p_{TE}(\infty)\ p_{TC}(\infty)\right]^T$. The terms $p_{TC}(\infty)$ and $p_{TE}(\infty)$ refer to the states TC and TE, respectively, when $(t \to \infty)$. The growth rate (r_g) can be given by the net transition rate from VS to MS_{c4} of the Markov chain shown in Figure 4.

$$r_g = \frac{1}{1/(r_{fi}-r_{r,ti2})+2/(k_bmi)+1/(k_{dc}-k_{et})+1/(r_{rfo}-rr,to2)} - \frac{r_{r,2}}{4}$$

The error rate vs. growth rate for the enveloped tile assembly has been calculated applying above derivations. Figure 5 provides an analysis of the error rate vs. growth rate for the enveloped tile, the original tile set and the 2x2 proofreading tile set (data from [10]). The original tile and 2x2 tile are represented by two curves corresponding to G_{se} = 6.5, 8.5 and G_{se} = 6.5, 7.5, respectively. The G_{mc} for these plots vary according to the relation (G_{mc}= 2 G_{se} - ϵ) while keeping G_{se} constant. For the enveloped tile simulations, two curves represent $G_{se} = 7.5$ and $G_{se} = 8.5$, respectively.

The optimum performance (error rate vs. growth rate) of each tile type is achieved along their respective slope $\frac{d[log(error\ rate)}{d(log(growth\ rate)}$ lines. Further, these slopes may be used to derive relations between growth rate (r_g) and error rate(ε). For enveloped tile, this relation is $r_g \approx \beta_{et}\varepsilon^{0.7}$ where $\beta_{et} \approx 8.25 \times 10^4/M/sec$. Error rate and growth rate relations for original tile, 2x2 tile, and Layered and Protected tiles of [4] are: Original Tile: $r_g \approx \beta_{original}\varepsilon^2$ ($\beta_{original} \approx 1 \times 10^4/M/sec$); 2x2 Tile: $r_g \approx \beta_{2x2tile}\varepsilon^{1.4}$ ($\beta_{2x2tile} \approx 1 \times 10^4/M/sec$); Protected Tile: $r_g \approx \beta_{PTM}\varepsilon^{1.4}$ ($\beta_{PTM} \approx 4.4 \times 10^2/M/sec$); Layered Tile: $r_g \approx \beta_{LTM}\varepsilon^{0.7}$ ($\beta_{LayeredTile} \approx 3.6 \times 10^2/M/sec$). Thus, the enveloped tile are more robust against errors than the original tile and 2x2 tiles whilst comparable to the Layered Tile.

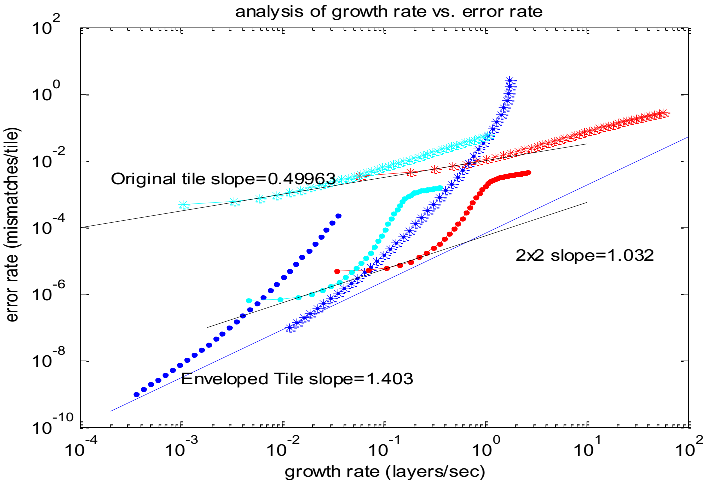

Fig. 5. Analytically derived growth rate vs. error rate plots of Original Tile (G_{se}=6.5, 8.5 (top curves)), 2x2 tile (G_{se}=6.5, 7.5 (middle curves)) and Enveloped Tile (G_{se}=7.5, 8.5 (bottom curves))

6 Conclusion and Future Work

A new Enveloped Tile approach for reliable tile assembly has been presented, together with an analysis of the involved association and dissociation processes necessary for correct tile assembly. The design of a complete Enveloped Tile set for the Siepinski Triangle assembly was presented, as an example of an application of the new approach. Error resilience of the enveloped tile assembly has been analysed using a physically realistic kinetic trapping model. Error rate vs. growth rate assessment revealed that the enveloped tile's error prevention capability is equivalent to (if not better than) the best case of the Layered tile.

It is anticipated that the fully protected output of Enveloped Tiles may in addition provide better prevention of other errors, i.e. facet nucleation and spontaneous nucleation errors, which warrants thorough analysis in further work. Other features that remain to be investigated include the impact of coiling, stiffness and swivelling of the central clamp of the PT in order to achieve a more accurate modelling of enveloped tile-mediated self-assembly.

The self-triggered activation principle aimed to guide an error-free tile assembly process may provide a general approach to designing molecular systems for DNA computing, synthetic biology and nanotechnology, where different components are often required to co-exist in order to allow conditional interactions and therefore be able to interact also illegitimately.

References

1. A., W.P., van der, H.T., Fernando, M.H., pakowitz Andrew S, Rob, P., Jonathan, W., Cees, D., Nelson, P.C.: : High flexibility of dna on short length scales probed by atomic force microscopy. Nature 2(1), 1748–3387 (2006)
2. Bhalla, N., Bentley, P., Vize, P., Jacob, C.: Staging the self-assembly process using morphological information. In: European Conf. on Artificial Life (ECAL), pp. 93–100 (2011)
3. Chen, H.L., Schulman, R., Goel, A., Winfree, E.: Reducing facet nucleation during algorithmic self-assembly. Nano Lett. 7(9), 2913–2919 (2007)
4. Fujibayashi, K., Zhang, D.Y., Winfree, E., Murata, S.: Error suppression mechanisms for dna tile self-assembly and their simulation. Natural Computing 8(3), 589–612 (2009)
5. Chen, H.-L., Goel, A.: Error free self-assembly using error prone tiles. In: Ferretti, C., Mauri, G., Zandron, C. (eds.) DNA10. LNCS, vol. 3384, pp. 62–75. Springer, Heidelberg (2005)
6. Majumder, U., LaBean, T.H., Reif, J.H.: Activatable tiles: Compact, robust programmable assembly and other applications. In: Garzon, M.H., Yan, H. (eds.) DNA 13. LNCS, vol. 4848, pp. 15–25. Springer, Heidelberg (2008)
7. Panyutin, I., Hsieh, P.: The kinetics of spontaneous dna branch migration. PNAS 91, 2021–2025 (1994)
8. Schulman, R., Winfree, E.: Programmable control of nucleation for algorithmic self-assembly. In: Ferretti, C., Mauri, G., Zandron, C. (eds.) DNA 10. LNCS, vol. 3384, pp. 319–328. Springer, Heidelberg (2005)
9. Winfree, E.: On the computational power of dna annealing and ligation. In: DNA Based Computers. DIMACS, vol. 27. American Mathematical Society (1995)
10. Winfree, E., Bekbolatov, R.: Proofreading tile sets: Error correction for algorithmic self-assembly. In: Chen, J., Reif, J.H. (eds.) DNA 9. LNCS, vol. 2943, pp. 126–144. Springer, Heidelberg (2004)
11. Winfree, E., Liu, F., Wenzler, L.A., Seeman, N.: Design and self-assembly of two-dimensional dna crystals. Nature 394(6693), 539–544 (1998)
12. Woo, S., Rothemund, P.W.K.: Programmable molecular recognition based on the geometry of dna nanostructures. Nat. Chem. 3(8), 620–627 (2011)